1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-like lymphoid aggregates in localized prostate tumors are not consistently associated with improved therapy response when tumors co-express high PTGER4 and high PD-L1. The PGE2–PTGER4 signaling axis amplifies PD-L1–driven T-cell exhaustion and impairs TLS maturation, preventing functional TLS-mediated anti-tumor activity. Specifically, PTGER4hi/PD-L1hi TLS-rich tumors will exhibit reduced objective responses to standard therapies (including PD-(L)1 inhibitors) due to (a) maintenance of exhausted CD8+ T cells (elevated PD-1, TIM-3, LAG-3, TOX; diminished IFN-γ, granzyme B), (b) impaired germinal center–like maturation within TLS (lower CXCL13, BCL6+ B cells, follicular dendritic cell networks), and (c) a suppressive cytokine milieu that sustains a non-responsive microenvironment. Consequently, blockade of PTGER4 signaling or PGE2 synthesis in combination with PD-(L)1 pathway inhibition will restore TLS maturation and T-cell functionality, improving treatment efficacy. Systemic inflammatory status (CRP, IL-6), dietary inflammatory indices (EDIH/EDIP), and metabolic signals (insulin/metabolite profiles) will modulate the PTGER4–PD-L1 axis and related imaging phenotypes, acting as contextual modifiers of response and helping to identify patients most likely to benefit from combination strategies. Predictions: (i) TLS-rich, PTGER4hi/PD-L1hi tumors show poorer response to monotherapy compared with PTGER4lo/PD-L1lo TLS-rich tumors; (ii) PTGER4 antagonism or PGE2 synthesis inhibition reshapes TLS architecture toward maturation and increases functional T-cell readouts; (iii) combined PTGER4/PGE2 blockade with PD-(L)1 blockade yields higher response rates in TLS-rich contexts; (iv) radiomic/pathomic features will correlate with PTGER4/PD-L1 status and predict response when integrated with systemic inflammatory/metabolic indices.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PTGER4, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTGER4: PTGER4–PD-L1–T-cell exhaustion axis) and CORE PATH 3 (imaging/therapy interplay linking infiltrate status to therapy effects), because they provide a mechanistic route whereby prostaglandin signaling can amplify checkpoint-driven exhaustion and plausibly disable TLS function detectable by imaging.

PD-L1: PD-L1–T-cell exhaustion axis) and CORE PATH 3 (imaging/therapy interplay linking infiltrate status to therapy effects), because they provide a mechanistic route whereby prostaglandin signaling can amplify checkpoint-driven exhaustion and plausibly disable TLS function detectable by imaging.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	T-cell exhaustion (functional markers)	Supporting modifier from knowledge graph
Secondary	EDIH/EDIP (dietary inflammatory indices)	Supporting modifier from knowledge graph
Secondary	CRP	Supporting modifier from knowledge graph
Secondary	IL6	Supporting modifier from knowledge graph
Secondary	insulin_response_markers	Supporting modifier from knowledge graph
Secondary	metabolite_biomarkers	Supporting modifier from knowledge graph
Secondary	radiomics/pathomics features	Supporting modifier from knowledge graph
Secondary	Olaparib (PARP inhibitor context)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- T-cell exhaustion (functional markers): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EDIH/EDIP (dietary inflammatory indices): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.